

LLM AGENTS · PLANNING SYSTEMS · SELF-AWARENESS

KnowSelf: Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

Agentic knowledgeable self-awareness: learn situational tokens that regulate direct actions, self-reflection, and external knowledge retrieval during interactive planning.

Source: [arXiv:2504.03553](#)

Focus: selective knowledge usage

ALFWorld benchmark

ONE-LINER

KnowSelf enables agents to autonomously decide when to act directly, when to reflect, and when to consult external knowledge—avoiding indiscriminate “flood irrigation.”

Audience: ML researchers & graduate students interested in LLM-based agents and planning systems.

PROBLEM

Current agent training uses “flood irrigation” – indiscriminate knowledge injection hurts efficiency

TRAJECTORY IMITATION

ReAct-style learning: fit gold trajectories even when the agent could have acted directly—risking fragile pattern fitting.

TRIAL-AND-ERROR REFINEMENT

Reflexion-style loops: external feedback is injected regardless of whether reflection is actually needed at a step.

KNOWLEDGE-AUGMENTED PLANNING

KnowAgent / WKM: consult knowledge on **100%** of steps—unnecessarily increasing inference cost and sometimes hurting performance.

- Key mismatch: injection is **situationally blind**—it ignores whether the agent can already solve the step.
- Observed consequences: fragile planning, pattern collapse on unexpected inputs, and high inference costs.
- Human contrast: metacognitive self-awareness—“can I act now, should I reflect, or do I need outside help?”

KNOWSELF TAXONOMY

Three situational modes: **fast**, **slow**, and **knowledgeable** thinking

DECISION CRITERION (PER STEP)

Compare predicted action to the gold action; if wrong, test whether **reflection** can self-correct; otherwise, require **external knowledge**.

→ Route to one of three modes

FAST THINKING

FAST · direct

Agent can directly produce the correct action with minimal deliberation.

SLOW THINKING

SLOW · reflect

Agent initially produces an incorrect action but self-corrects through reflection.

KNOWLEDGEABLE THINKING

KNOW · retrieve

Agent cannot produce the correct action even after reflection and must consult external knowledge.

KnowSelf learns to generate special situation tokens at inference time that trigger **direct action**, **reflection**, or **knowledge retrieval**.

METHOD

KnowSelf framework: data construction → two-stage training → self-aware inference

END-TO-END PIPELINE (FIGURE 3)

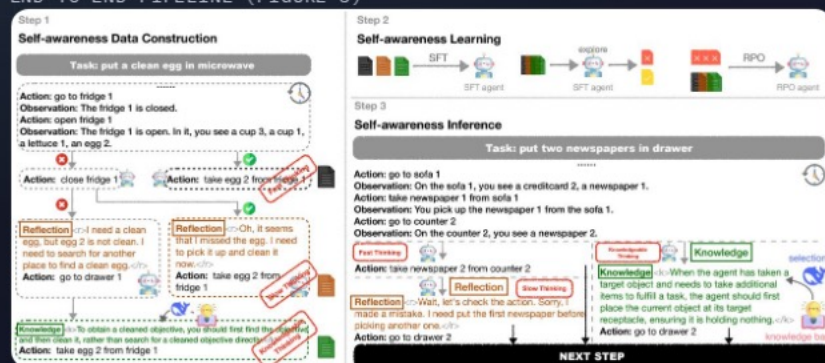


Figure: data construction, learning, and inference pipeline.

STEP 1 – SELF-AWARENESS DATA CONSTRUCTION

Self-explore trajectories; heuristically label each step: match gold → **fast**; reflection fixes → **slow**; else → **knowledgeable**. Insert special tokens.

STEP 2 – TWO-STAGE TRAINING

Train with **SFT** to learn initial token patterns, then refine with **RPO** to improve token generation quality.

STEP 3 – SELF-AWARE INFERENCE

At deployment, the agent generates tokens that selectively trigger **reflection** or **knowledge retrieval**—reducing unnecessary calls.

QUALITATIVE

KnowSelf correctly heats a mug and places a saltshaker where baselines fail

FIGURE 1

(a) Task: put a hot mug in cabinet	
OpenAI-01 Action: take mug 1 from countertop 1 Observation: You pick up the mug 1 from the countertop 1. Action: go to microwave 1 Observation: The microwave 1 is closed. Action: open microwave 1 Observation: You open the microwave 1. In it, you see an egg 1. [Knowledge] <- If the agent is in a situation where it needs to modify the temperature of an object using a receptacle (e.g., a fridge for cooling), it should directly perform the specific action like 'cool' or 'heat' instead of placing the object on it.> Action: take egg 1 from microwave 1 Observation: Nothing happens.	KnowSelf (Llama-8B) Action: take mug 1 from countertop 1 Observation: You pick up the mug 1 from the countertop 1. Action: go to microwave 1 Observation: The microwave 1 is closed. Action: open microwave 1 Observation: You open the microwave 1. In it, you see an egg 1. Action: put mug 1 in/on microwave 1 [Reflection] <- Wait, let's check the action. Let's think step by step. With the microwave open, I can heat the mug using the microwave.> Action: heat mug 1 with microwave 1
(b) Task: put a saltshaker in drawer	
DeepSeek-R1 Action: take saltshaker 1 from countertop 3 Observation: You pick up the saltshaker 1 from the countertop 3. Action: go to drawer 1 Observation: The drawer 1 is open. In it, you see a fork 1. Action: put saltshaker 1 in/on drawer 1 [Reflection] <- Wait, the drawer is open, so the saltshaker might fall inside. I need to close the drawer first.> Action: close drawer 1 Observation: You close the drawer 1. Action: put saltshaker 1 in/on drawer 1 Observation: Nothing happens.	KnowSelf (Llama-8B) Action: take saltshaker 1 from countertop 3 Observation: You pick up the saltshaker 1 from the countertop 3. Action: go to drawer 1 Observation: The drawer 1 is closed. Action: open drawer 1 Observation: You open the drawer 1. In it, you see a fork 1. [Knowledge] <- When the agent encounters a situation where receptacles contain other items but only needs to place the target object, the agent should proceed to place the object without removing unrelated items.> Action: put saltshaker 1 in/on drawer 1

Figure: KnowSelf vs. OpenAI-01 and DeepSeek-R1 on ALFWorld tasks.

TASK A – “PUT A HOT MUG IN CABINET”

OpenAI-01 opens microwave then incorrectly tries to take an egg; **KnowSelf (Llama-8B)** emits a **Reflection** token and realizes it should heat the mug using the open microwave.

TASK B – “PUT A SALTSHAKER IN DRAWER”

DeepSeek-R1 reflects and attempts to close the drawer first (fails); **KnowSelf** applies a **Knowledge** token noting it should place the object without removing unrelated items.

Interpretation: situational tokens gate *which* cognitive resource to spend—reflection vs retrieval—rather than always invoking both.

RESULTS (ALFWORLD)

KnowSelf achieves **84.33%** on ALFWorld with Llama-8B using only **15%** knowledge

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

Know% = proportion of steps that query external knowledge. Full-knowledge methods (ExpeL / KnowAgent / WKM) use 100%.

EFFICIENCY

Minimal knowledge, maximum impact: efficiency advantage over full-knowledge methods



READING THE CHART

Accuracy (All) remains high while **knowledge usage** drops from **100%** to **~15–26%**.

Scaling: even **Gemma-2B** reaches **79.85%** with selective knowledge—matching GPT-4o Expel’s overall score in the table.

■ All accuracy (%) ■ Know% (knowledge-query steps)

Bars are normalized to 100 for visual comparison; values are exact from the results table.

TAKEAWAYS

Key takeaways: self-awareness is the missing piece in agent planning

- **Flood irrigation is wasteful.** Indiscriminate injection of trajectories, feedback, or knowledge increases cost and can degrade robustness.
- **Situational tokens enable metacognition.** Lightweight heuristic labeling on self-explored trajectories marks **fast**, **slow**, and **knowledgeable** situations.
- **Strong empirical results.** KnowSelf (Llama-8B) reaches **84.33%** overall, exceeding GPT-4o ReAct (76.12) and GPT-4o Expel (79.85) in the table.
- **Efficiency wins.** Only **15.01%** (Llama-8B) and **26.41%** (Gemma-2B) steps use knowledge vs **100%** for full-knowledge baselines.
- **Generalizes across scale.** Gemma-2B with KnowSelf reaches **79.85%** overall—matching GPT-4o Expel’s score while using far less knowledge.

FUTURE DIRECTIONS (FROM THE FRAMING)

Push situational self-awareness beyond ALFWorld: richer knowledge sources, more diverse environments, and tighter coupling between planning and retrieval—while keeping knowledge usage selective rather than default.